

Unit 14, Lesson 4: Comparing Theoretical Probability to Experimental Probability

Benchmarks

MA.8.DP.2.1 Determine the sample space for a repeated experiment.

MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment.

Learning Target

- ☐ I can calculate theoretical probability and compare it to experimental probability to evaluate accuracy.

14.4.1 Warm-Up: Data Analyst

Ivana Seric has her PhD in mathematics. She works as a data analyst for the Philadelphia 76ers, an NBA basketball team. Data science has changed the game of basketball as players and coaches have been able to use data and analytics, rather than instincts and talent alone, to determine how to improve the game. Data analytics is a growing career field in a world where more and more data are collected and shared through increasing technological means.

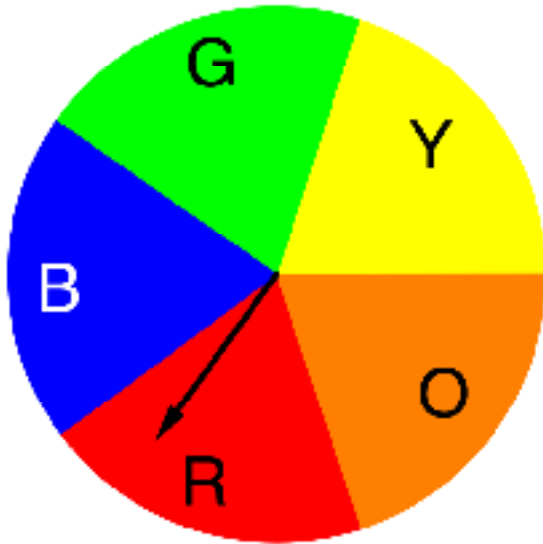


1. What are some ways data analytics has changed the game of basketball?
2. Explain how other industries might use statistics.

14.4.2 Exploration: Calculating Theoretical Probability

Work with your partner to complete the following simulation involving spinning Spinner B and flipping a fair coin.

Spinner B

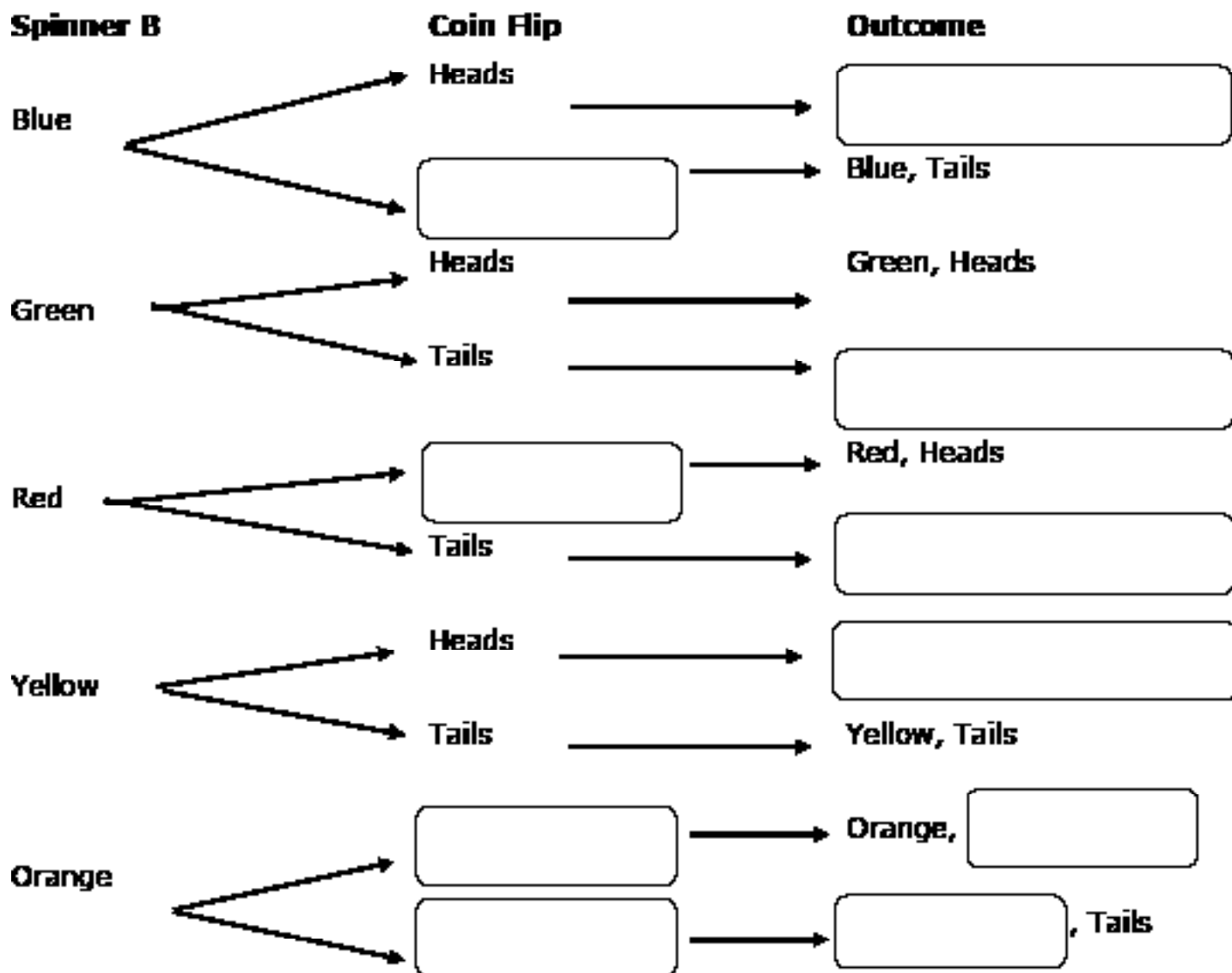


Coin



As part of a game, Spinner B is spun and a coin is flipped. The theoretical probabilities of events in this random process can be found by determining the sample space using a list, table, or tree diagram.

1. Complete the tree diagram to determine the sample space.



2. Find the theoretical probability that the spinner lands on yellow and heads is flipped on the coin.

$$\begin{aligned} P(\text{yellow, heads}) &= \\ &\frac{\text{the number of outcomes that are (yellow, heads)}}{\text{total number of outcomes in the tree diagram}} \\ &= \frac{1}{\boxed{}} = 0.100 = \boxed{} \% \end{aligned}$$

3. Find the theoretical probability that the spinner *does not* land on blue and the coin lands tails up, $P(\text{not blue, tails})$.
4. What is the theoretical probability that the spinner lands on a primary color (red, blue, or yellow) and the coin lands tails up, $P(\text{primary color, tails})$?

One way to evaluate the accuracy of theoretical probabilities is to compare them to experimental probabilities.

- 5. Work with your partner to conduct 50 trials of the game where Spinner B is spun and a fair coin is flipped. Using a fair decision-making process, decide which partner will be in charge of spinning the spinner and which will be in charge of flipping the coin.**

Record your data in the table by tallying each outcome in the middle column. After you have completed the simulation 50 times, record the total number of occurrences for each outcome.

Outcome	Number of Times the Outcome Occurs	Experimental Probability of Each Outcome
Red, heads	Total:	
Red, tails	Total:	
Orange, heads	Total:	
Orange, tails	Total:	
Yellow, heads	Total:	

Outcome	Number of Times the Outcome Occurs	Experimental Probability of Each Outcome
Yellow, tails	Total:	
Green, heads	Total:	
Green, tails	Total:	
Blue, heads	Total:	
Blue, tails	Total:	
Overall Total	50	1 = 100%

6. Find the experimental probability of each outcome in the sample space and record it in the last column in the table. For example, to find the experimental probability of the outcome "yellow, heads," divide the number of times the outcome occurred by the total number of trials. Suppose "yellow, heads" occurred 7 times in the experiment.

$$P(\text{yellow, heads}) = \frac{7}{50} = \frac{\boxed{}}{100} = \boxed{} = 14\%$$

7. Find the experimental probability that the spinner *does not* land on blue and the coin lands tails up, $P(\text{not blue, tails})$.

8. What is the experimental probability that the spinner lands on a primary color (red, blue, or yellow) and the coin lands tails up, $P(\text{primary color, tails})$?

9. Explain how these experimental probabilities compare to the theoretical probabilities you calculated using the tree diagram.

10. Why do you think the experimental probabilities differed from the theoretical probabilities?

14.4.3 Collaboration: Compound Probability

1. With your partner, complete the probability table to show the theoretical probabilities of each step of an experiment and the overall probability of the event.
- For example, in the P(yellow, heads) row, the theoretical probability of the first step of this event, spinning yellow on Spinner B, is $\frac{1}{5}$; therefore, $\frac{1}{5}$ is written in the "...First step" column. Next to that, the theoretical probability of landing on heads when a fair coin is flipped, $\frac{1}{2}$, is recorded. The overall theoretical probability, P(yellow, heads), $\frac{1}{10}$, is written in the "...Event" column.

Event		Theoretical Probabilities of the...		
		...First Step	...Second Step	...Event
Spin Spinner B, Flip a Coin	P(yellow, heads)	$\frac{1}{5}$	$\frac{1}{2}$	$\frac{1}{10}$
	P(not blue, tails)	$\frac{4}{5}$		
	P(primary color, tails)		$\frac{1}{2}$	$\frac{3}{10}$

2. Suppose a fair coin is tossed three times. Complete the table.

Event	Theoretical Probability of ...			
	First Toss is Heads	Second Toss is Heads	Third Toss is Heads	All Three Tosses are Heads
Flip three heads in a row	$\frac{1}{2}$			$\frac{1}{8}$

3. What do you notice about the tables?

4. What do you wonder about the tables?

14.4.4 Bring It Together: Calculating Theoretical Probability

- The theoretical probability of the outcomes in a multi-step experiment can be found by
 - adding
 - multiplying
 the
 - theoretical probability
 - experimental probability
 of the outcome for each step of the experiment.
- In a two-step experiment where a fair, six-sided die is rolled and a fair coin is flipped, the theoretical probability of rolling a 3 and the coin landing on tails, $P(3, \text{tail})$, is found by:

First, find the theoretical probability of getting a 3 on a fair

six-sided die, $P(3) =$

- $\frac{1}{6}$
- $\frac{1}{2}$

. Next, find the

- experimental
- theoretical

probability of a fair coin landing on

tails, $P(\text{tails}) =$

- 0.500.
- 0.250.

Finally,

- multiply
- add

the

probabilities of these steps to determine the probability of

the outcome, $\frac{1}{6} \cdot$

$= \frac{1}{12}$.

So, $P(3, \text{tails}) = \frac{1}{12} =$

$= 8.3\%$.

In a two-step experiment where the results in one step do not influence the results in the other step, $P(A \text{ and } B) = P(\text{event } A) \cdot P(\text{event } B)$.

3. In a repeated experiment where a fair coin is tossed six times, the theoretical probability of landing on tails every time is found by

- adding
 - multiplying

 the theoretical probability of landing on tails for each trial.

$P(\text{all tails}) =$

- $P\left(\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}\right) = \left(\frac{1}{2}\right)^6 = \frac{1}{64} = 0.015625 = 1.5625\%$
 - $2^6 = 64$.

14.4.5 Wrap-Up: Cool Down

1. **Write a short note to a friend who missed today's class summarizing what you learned.**

